

CSE 445: Machine Learning

Project Report Guide

Summer 2026

Your Machine Learning Project Title here

Using the following guidelines write your project report in IEEE conference format (see the other document)

1. Title Page

- **Project Title:** [Insert Title]
- **Project Group/Lead/Submitter:** [Insert Name]
- **Course/Section/Department:** [Insert Name]
- **Date:** [Insert Date]

2. Project Abstract (Summary)

Provide a brief summary of the project, including its purpose, key goals, and expected impact. This section should be concise and give an overview of what the project is about.

3. Project Introduction/Objectives/Motivations/Contributions

List the main objectives that the project aims to achieve. Make them clear, measurable, and achievable. Examples might include:

- **Objective 1:** [Specific objective related to the project]
- **Objective 2:** [Another objective]
- **Objective 3:** [Another objective]

4. Project Background

- **Context and Rationale:** Explain the background and context for the project. Why is it needed? What challenges or opportunities does it address?
- **Historical and Scientific Context:** Outline the historical development and scientific principles underlying the project's domain.
- **Significance to the Broader Domain:** Explain how the project fits within the larger context of the field and its potential impact on future developments.

5. Literature Review

- **Existing Knowledge and Theoretical Basis:** Summarize the current state of knowledge and theoretical frameworks related to the project. What concepts, models, or technologies form the foundation?
- **Gaps and Limitations in Previous Studies:** Identify the shortcomings, unresolved issues, or inconsistencies in the existing literature that justify the need for further investigation.
- **Relevant Case Studies or Applications:** Highlight significant case studies, experiments, or practical implementations from past research that demonstrate relevance and support the direction of the project.
- **Your project's relativity and advantage:** Demonstrate how project is related or different to existing body of research, how your project cover the limitations of the existing work, what new contributions/betterment you have implemented in your project

6. Design Methodology/Approach

- **Overview of Approach:** Provide a description of how the project is carried out.
- **Pipeline of the ML project:** Provide Data Preprocessing, Training and Testing pipeline in details.
- **System Architecture:** Provide a comprehensive figure that describes the system architecture of your project.
- **Tools and Resources:** Identify the tools, resources, or technologies that are used.

7. Testing Methodology

- **How Success Will Be Measured:** Define the criteria for testing the success of the project. This could include:
 - How the project is evaluated, what benchmarks will be used, how accuracy, precision or other parameters is measured
 - Completion on time and within budget
 - Achievement of specific objectives
 - Stakeholder satisfaction
- **Monitoring and Reporting:** Describe how the project is monitored and progress reported.

8. Project Timeline

Provide a timeline or Gantt chart with key milestones and deadlines:

- **Phase 1:** [Start Date – End Date]
- **Phase 2:** [Start Date – End Date]
- **Phase 3:** [Start Date – End Date]
- **Final Deliverable:** [Date]

9. Budget

Provide an estimated budget for the project, including all major costs such as:

- **Personnel:** [Cost for staff or contractors]
- **Materials/Equipment:** [Cost for necessary materials or equipment]
- **Software/Tools:** [Costs for any tools or software needed]
- **Miscellaneous:** [Other anticipated costs]
- **Total Budget:** [Total amount]

10. Team Members and Roles

List the key team members and their roles in the project. For example:

- **Project Manager:** [Name] – Responsible for overseeing the project's execution and timeline.
- **Team Member 1:** [Name] – Responsible for [specific tasks].
- **Team Member 2:** [Name] – Responsible for [specific tasks].

11. Risks and Mitigation Strategies

Identify potential risks to the success of the project and outline strategies to mitigate these risks.

- **Risk 1:** [Describe the risk]
 - Mitigation Strategy: [How to address the risk]
- **Risk 2:** [Describe the risk]
 - Mitigation Strategy: [How to address the risk]

12. Conclusion and Limitations

Summarize the key points of the proposal, emphasizing the value and potential impact of the project. Reinforce why this project should be considered as state-of-the-art and can be published in reputed conference and journal publications.

14. References

Provide any references or sources cited in the proposal, such as studies, reports, or articles that support your project.